

OVERSPEEDING SENSOR

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Problem Statement:

- **According to the NCRB, every year approximately 80,000 deaths are reported due to over speeding accidents.**
- **Over speed riding is one of the well-known scenarios we can see in youth.**
- **So, our main aim behind choosing this topic is to develop equipment which prevents accidents due to high speed.**
- **Also, there is lot of advancement in such equipment's used by transport department but here we tried to make this sensor completely with analog circuits and elements.**

Working of the Circuit:

- **Firstly, we will get trigger using IR Sensor/ LDR when a vehicle passes.**
- **Then we placed a differentiator circuit with PNP transistor.**
- **Getting trigger from sensors we fed them into two 555 timer IC which is further connected to NAND gate and output of NAND gate is given to other 555 timer.**
- **Using concepts of monostable and time for which trigger required, we have selected components and finally buzzer will ring to alert regarding over speeding.**
- **Also, we kept in mind about different breakdown values of elements used in this project like general purpose transistors(2N3906) have minimum emitter-base breakdown voltage of 5 volts, but we kept it below that to ensure smooth working of circuit.**

Challenges faced while solving this problem:

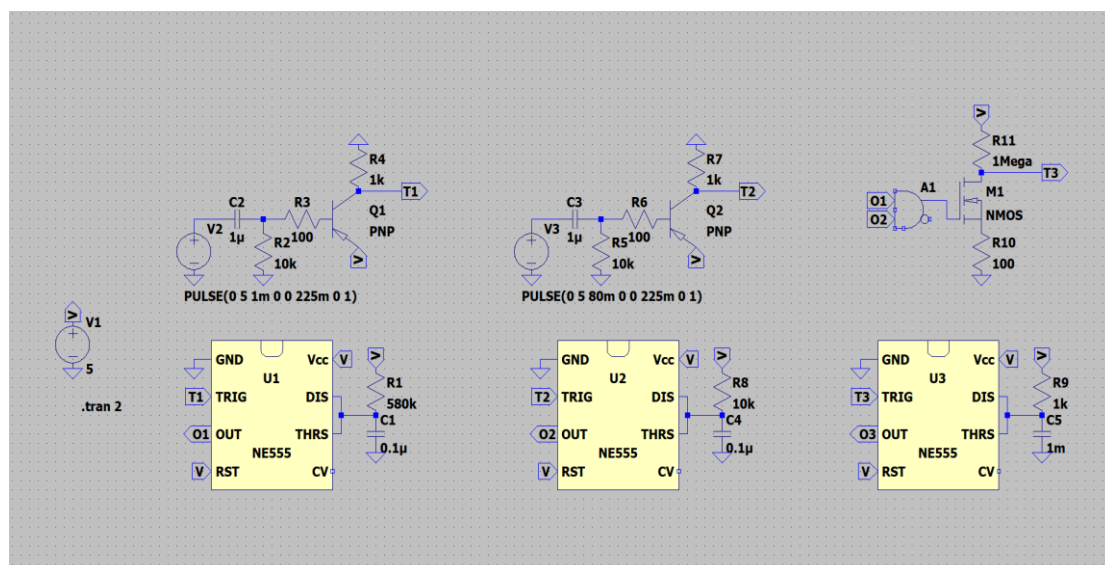
- What is the distance between 2 LDR/ IR sensors so that we must avoid overtaking?
- How to ensure that trigger only occurs when vehicle just enter in the field of sensors means trigger does not remain till the whole car is present in the field of sensors?

Solution of the problems stated:

- To prevent overtaking we tried to minimize the distance between two LDR as much as possible keeping the average length of the vehicles in mind.
- To solve 2nd problem, we put a differentiator circuit along with the NPN transistor to ensure that we will get trigger once when vehicle just cross the 1st Sensor/ LDR and then no trigger till passing through that LDR/ Sensor.

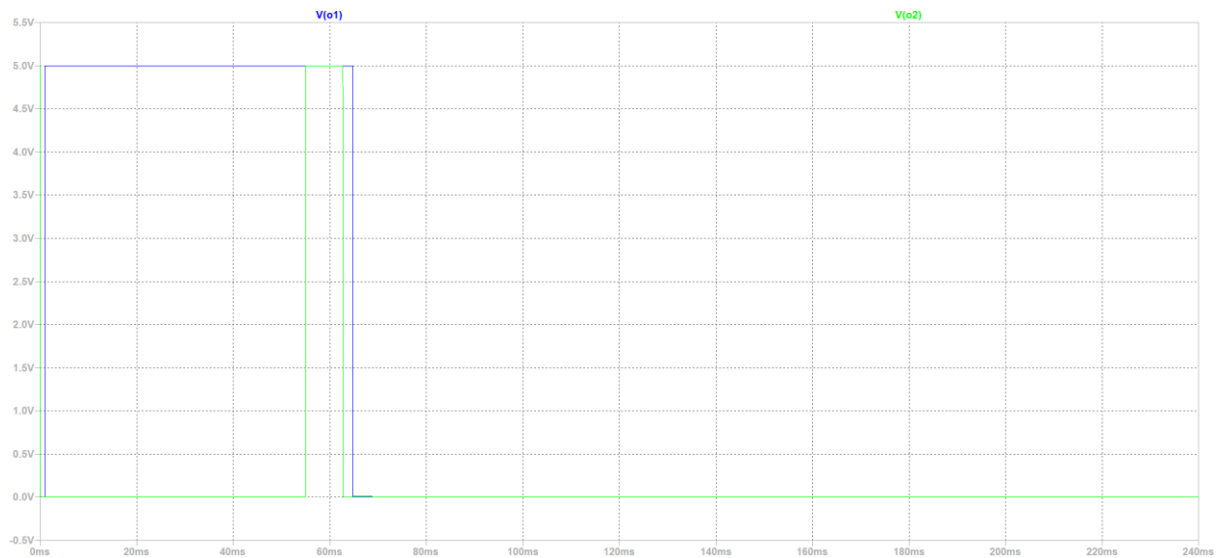
LTSPICE Diagram and Simulations:

Schematic:

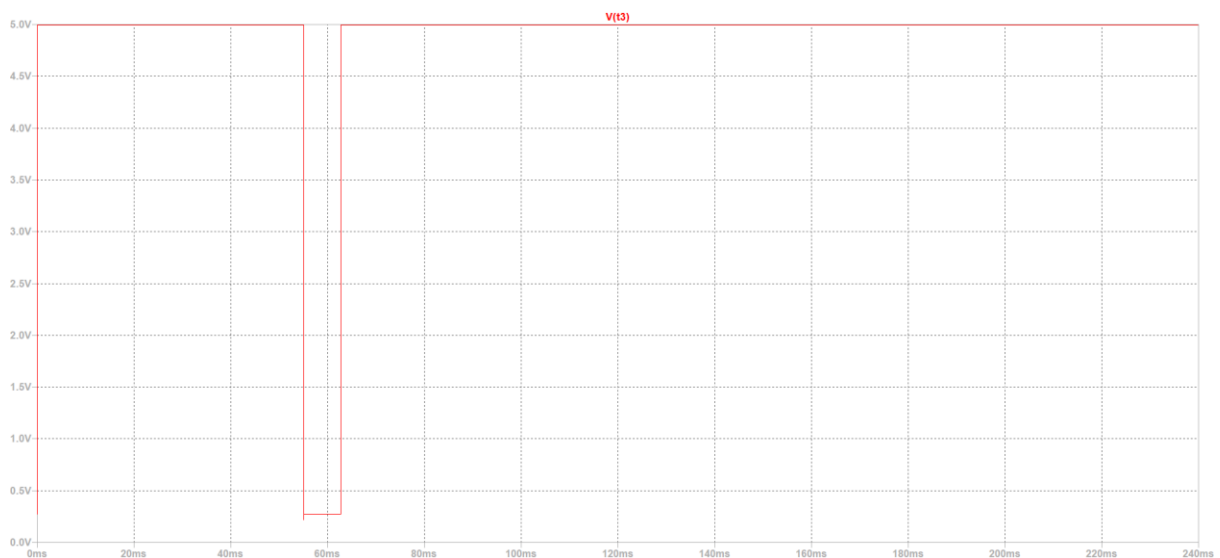


Simulations:

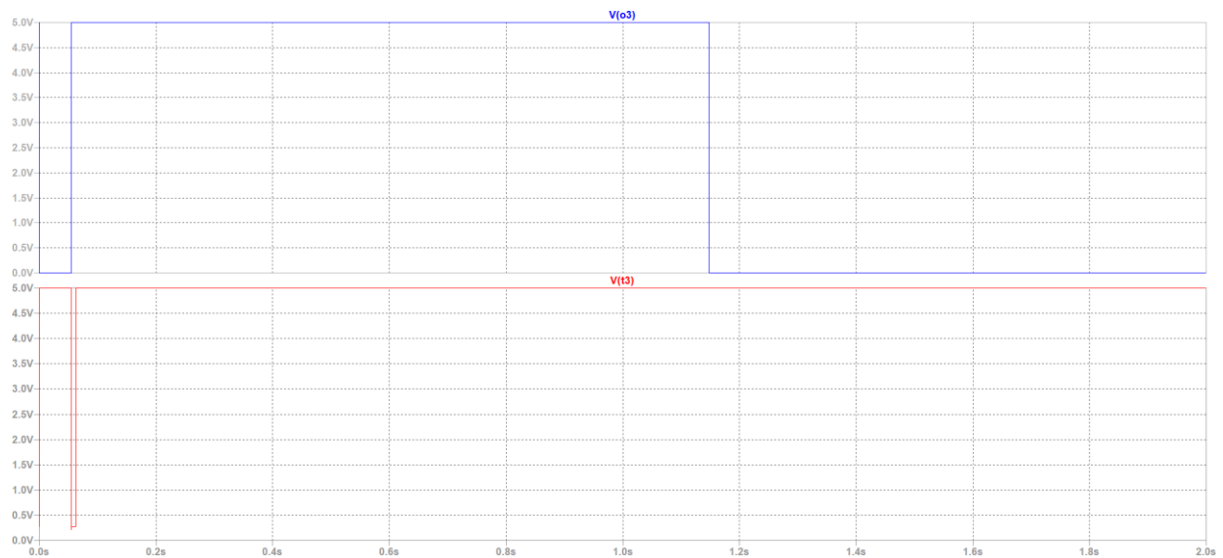
- Output voltage from the first two 555 timers that is trigger of 5 volts.



- Output from NAND gate: Since both triggers are high.

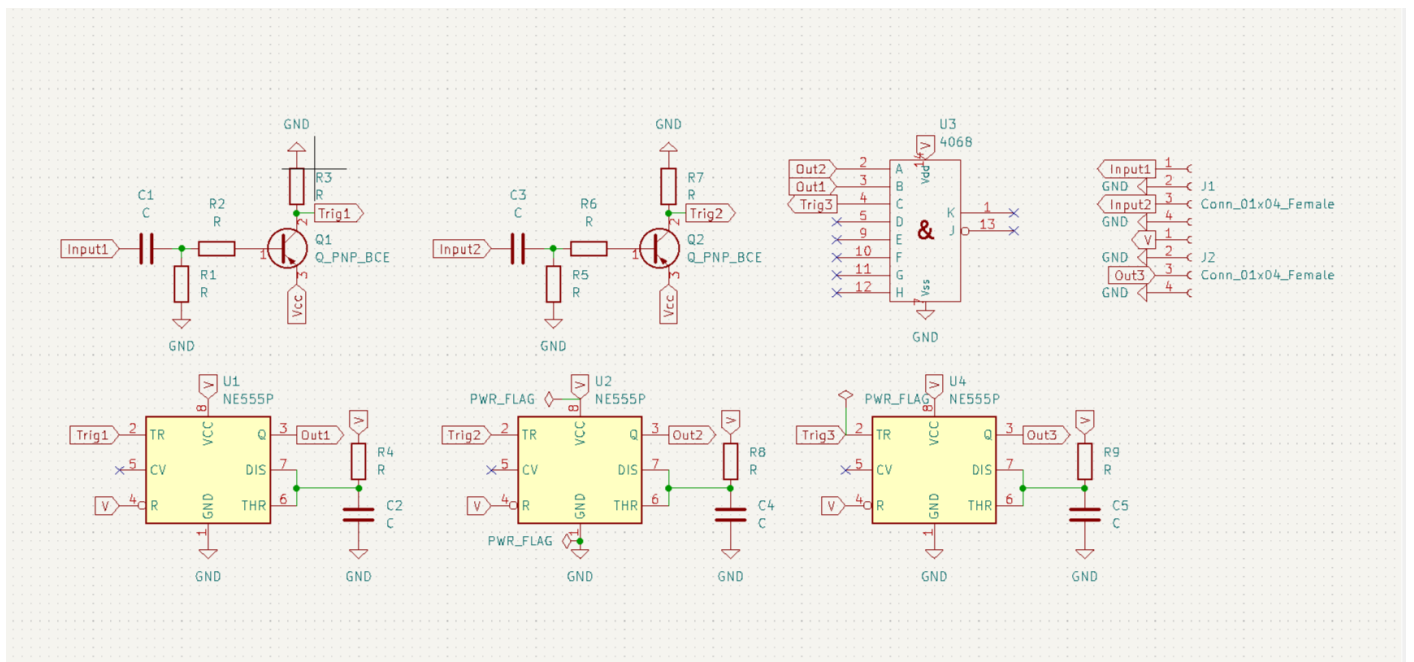


- Final Output from 3rd 555 timer IC



KiCAD Report: -

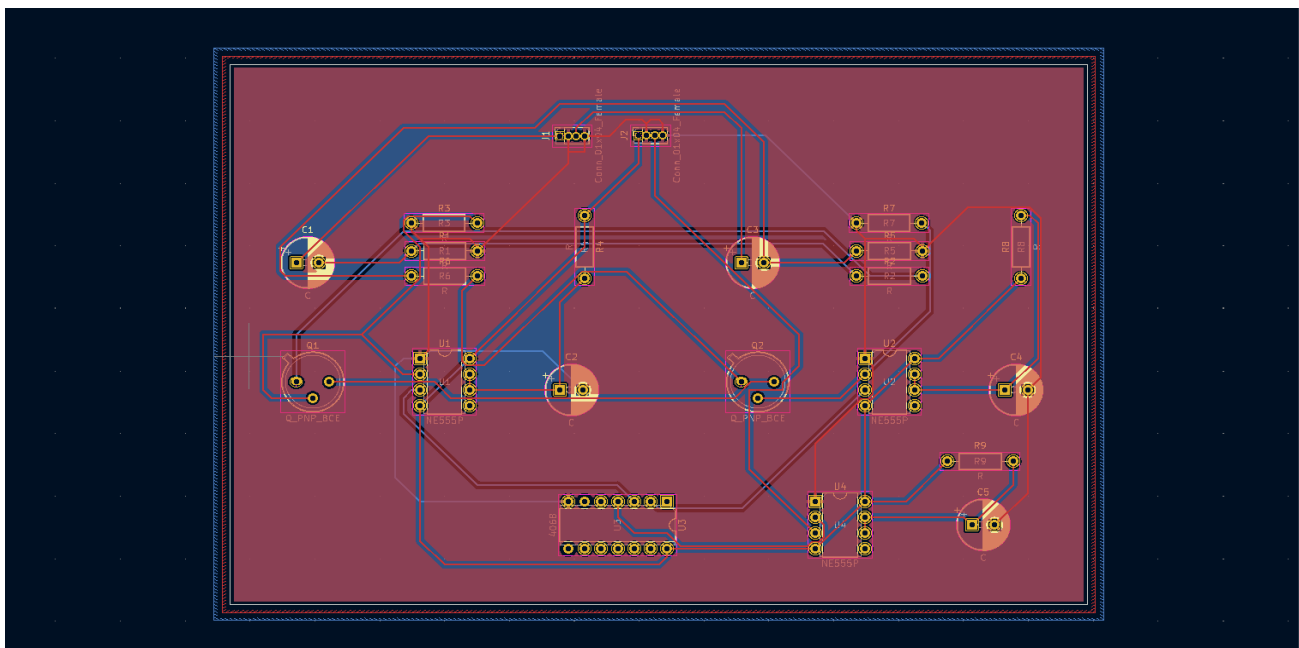
Schematic:



Footprint: Circuit elements we have used on PCB board.

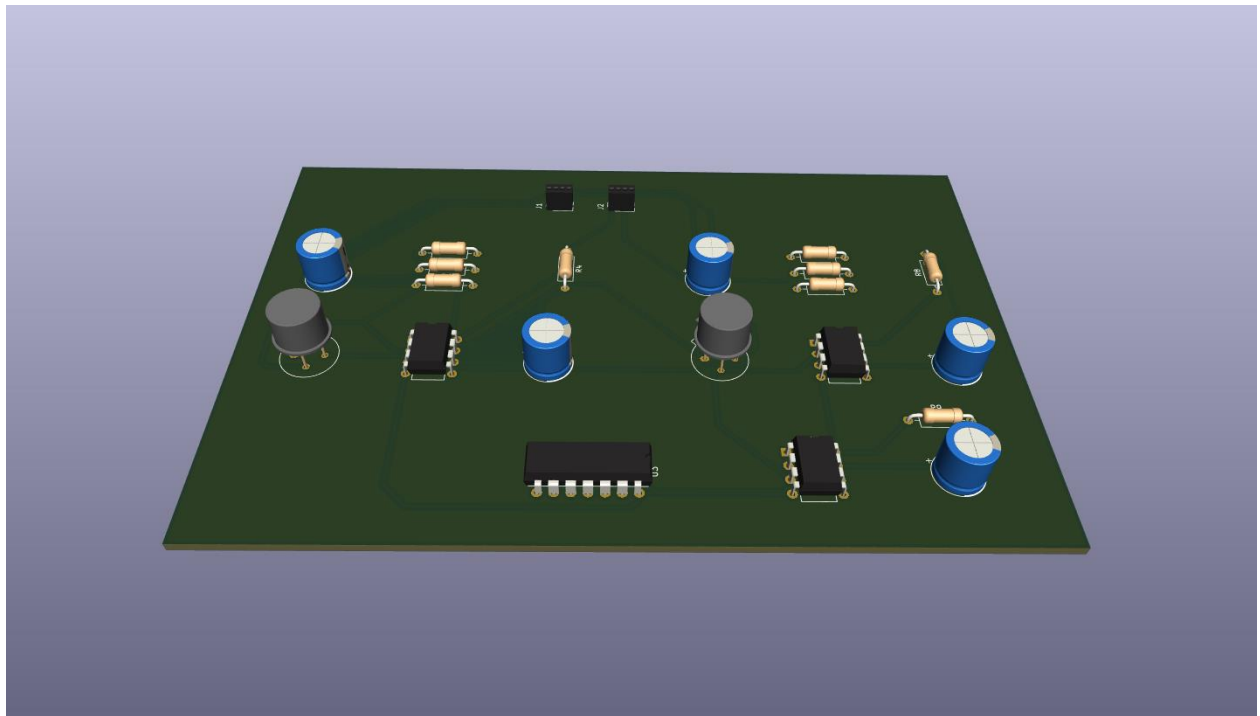
1	C1 -	C : Capacitor_THT:CP_Radial_D8.0mm_P3.50mm
2	C2 -	C : Capacitor_THT:CP_Radial_D8.0mm_P3.50mm
3	C3 -	C : Capacitor_THT:CP_Radial_D8.0mm_P3.50mm
4	C4 -	C : Capacitor_THT:CP_Radial_D8.0mm_P3.50mm
5	C5 -	C : Capacitor_THT:CP_Radial_D8.0mm_P3.50mm
6	J1 - Conn_01x04_Female :	Connector_PinSocket_1.27mm:PinSocket_1x04_P1.27mm_Vertical
7	J2 - Conn_01x04_Female :	Connector_PinSocket_1.27mm:PinSocket_1x04_P1.27mm_Vertical
8	Q1 -	Q_PNP_BCE : Package_TO_SOT_THT:TO-5-3
9	Q2 -	Q_PNP_BCE : Package_TO_SOT_THT:TO-5-3
10	R1 -	R : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
11	R2 -	R : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
12	R3 -	R : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
13	R4 -	R : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
14	R5 -	R : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
15	R6 -	R : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
16	R7 -	R : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
17	R8 -	R : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
18	R9 -	R : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
19	U1 -	NE555P : Package_DIP:DIP-8_W7.62mm
20	U2 -	NE555P : Package_DIP:DIP-8_W7.62mm
21	U3 -	4068 : Package_DIP:DIP-14_W7.62mm
22	U4 -	NE555P : Package_DIP:DIP-8_W7.62mm

PCB Layout:

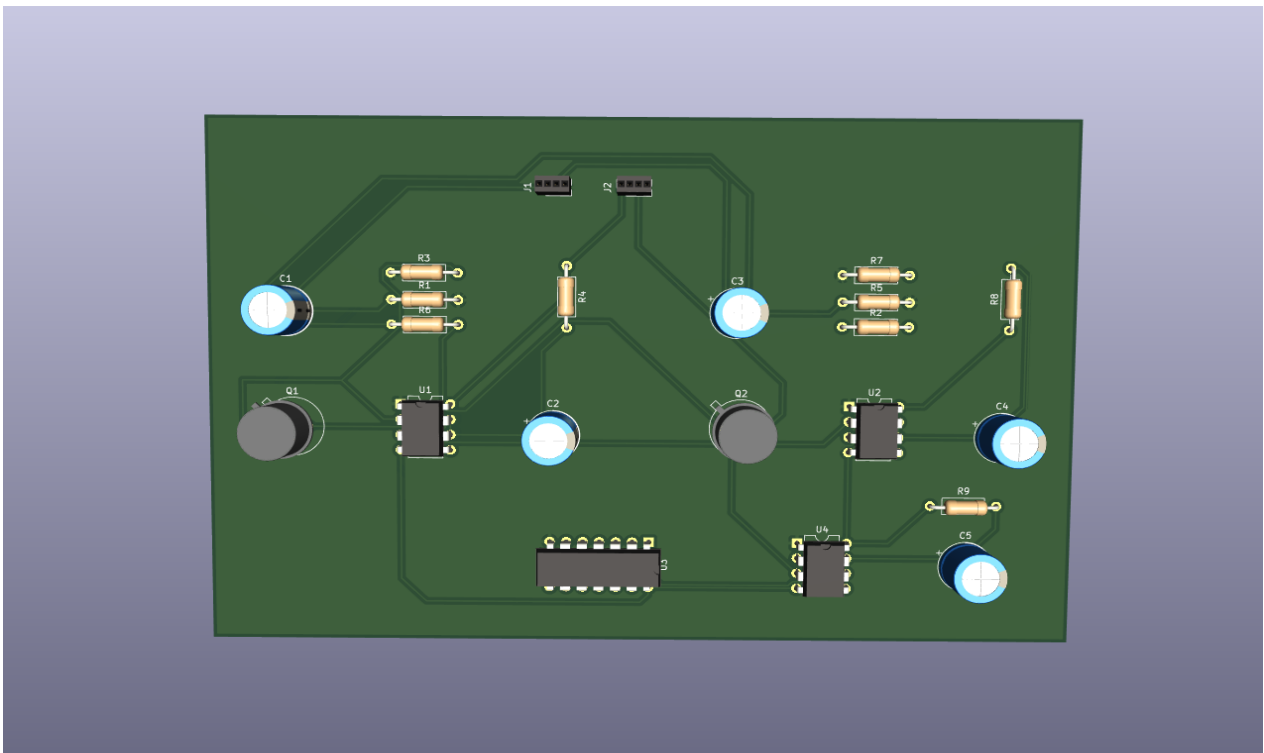


PCB MODEL:

Side View:



Front View:



BACK SIDE VIEW:

